

YXIZ: NeRF-Assisted & Deep Learning-Driven Spatial Inference for At-Home Scoliosis Detection

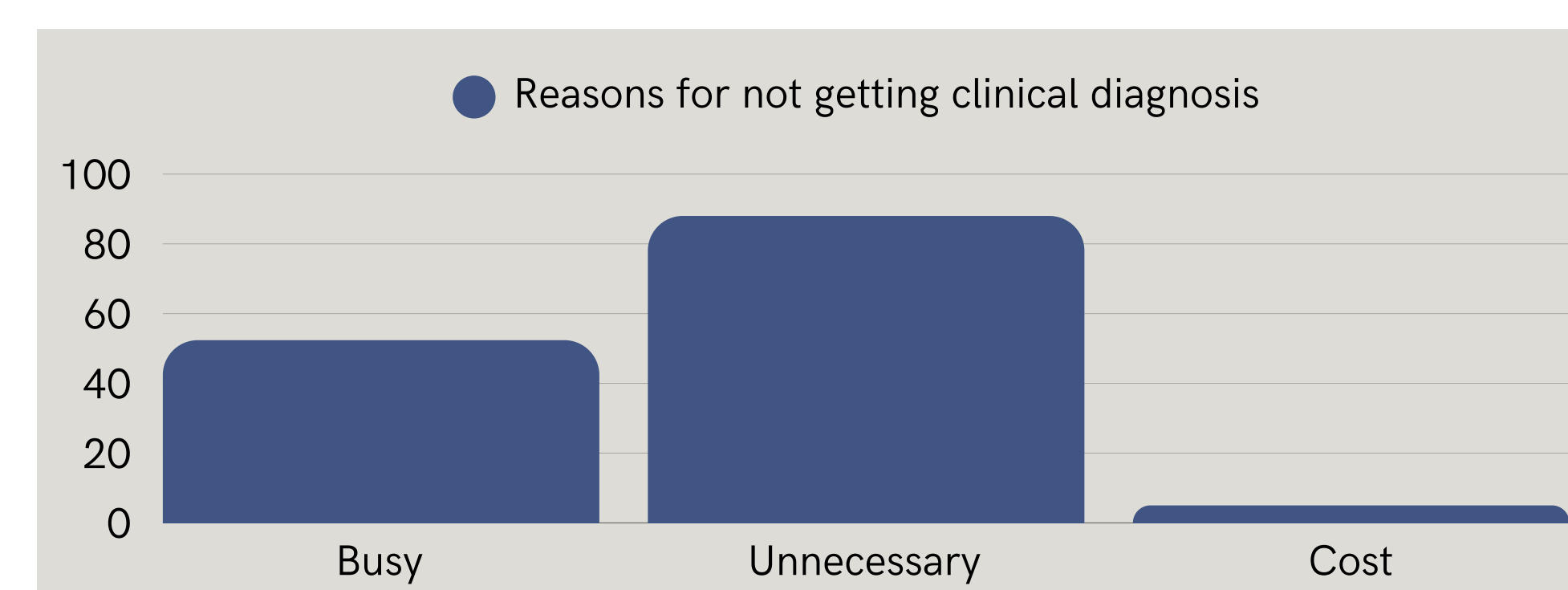
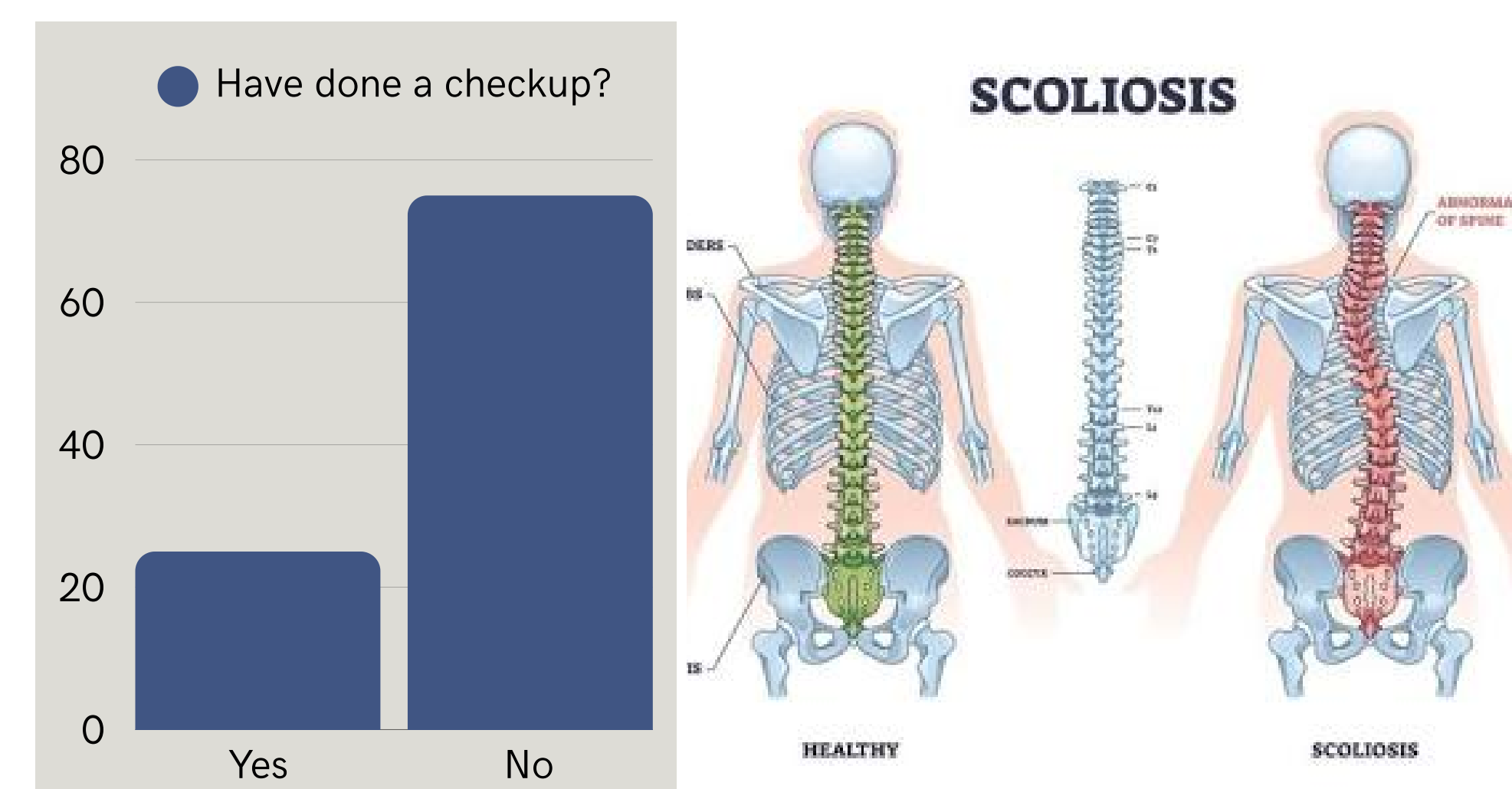
S. Yvette Chen

yvette_chen@berkeley.edu

Jacobs Innovation Catalyst

INTRODUCTION

Early scoliosis detection is crucial, yet current assessment methods rely on in-clinic, X-ray-based evaluations that are **intermittent and often inaccessible**. Many individuals—due to geographic distance, limited time, disability, cultural or religious constraints on imaging, or sensitivity to radiation—**cannot sustain consistent clinical visits**, leading to delayed diagnosis and avoidable progression.



YXiZ addresses this gap through a **digital, at-home screening pipeline** that **reconstructs a user's body geometry** from multi-view videos using NeRF-assisted computational imaging, followed by deep-learning-driven keypoint inference to approximate spinal curvature and **identify potential scoliosis patterns**.

By **reducing dependence on clinic availability and lowering the barrier to diagnostic understanding**, the project directly benefits populations historically underserved by traditional healthcare infrastructures, emphasizing autonomy, accessibility, and inclusivity across socioeconomic and cultural contexts.

ARCHITECTURES

The architecture includes three phases:

1. 3D Body **Reconstruction**,
2. Key point **prediction**, and
3. **Classification** of key point pattern into scoliosis category

Phase	Algorithm	Topology	Sensitivity	Complexity
Reconst.	Gaussian	volumetric	High	/
	NeRF	volumetric	High	/
	SMPL	standardized	Low	/
Predict.	GNN	standardized	/	Mid-Low
	mplicity GN	volumetric	/	High
Classif.	GNN	/	/	/

Bottleneck: Topology Standardization. The first bottleneck standardizing the input/output topology to fit the algorithms of three phases.

The easiest way would be using SMPL + GNN because the GNN directly lives on SMPL's standardized topology. However, SMPL as a parameter-based model inevitably overfits and standardizes spinal curvature, resulting in loss of information.

FUTURE WORK

Community Accessibility. In attempt to open-source the tool, there comes the need of an end-to-end software to make the project fully user-centric.

Data quality. Work that improves the efficiency and quality of labeling point cloud data would be greatly appreciated.

Tutorials. Enable interaction between patients and rehabilitation coaches who produce tutorials for specific types of scoliosis.

A major contribution of this project is thus providing a machine learning (ML) algorithm to standardize the topology of point clouds outputted by human NeRF.

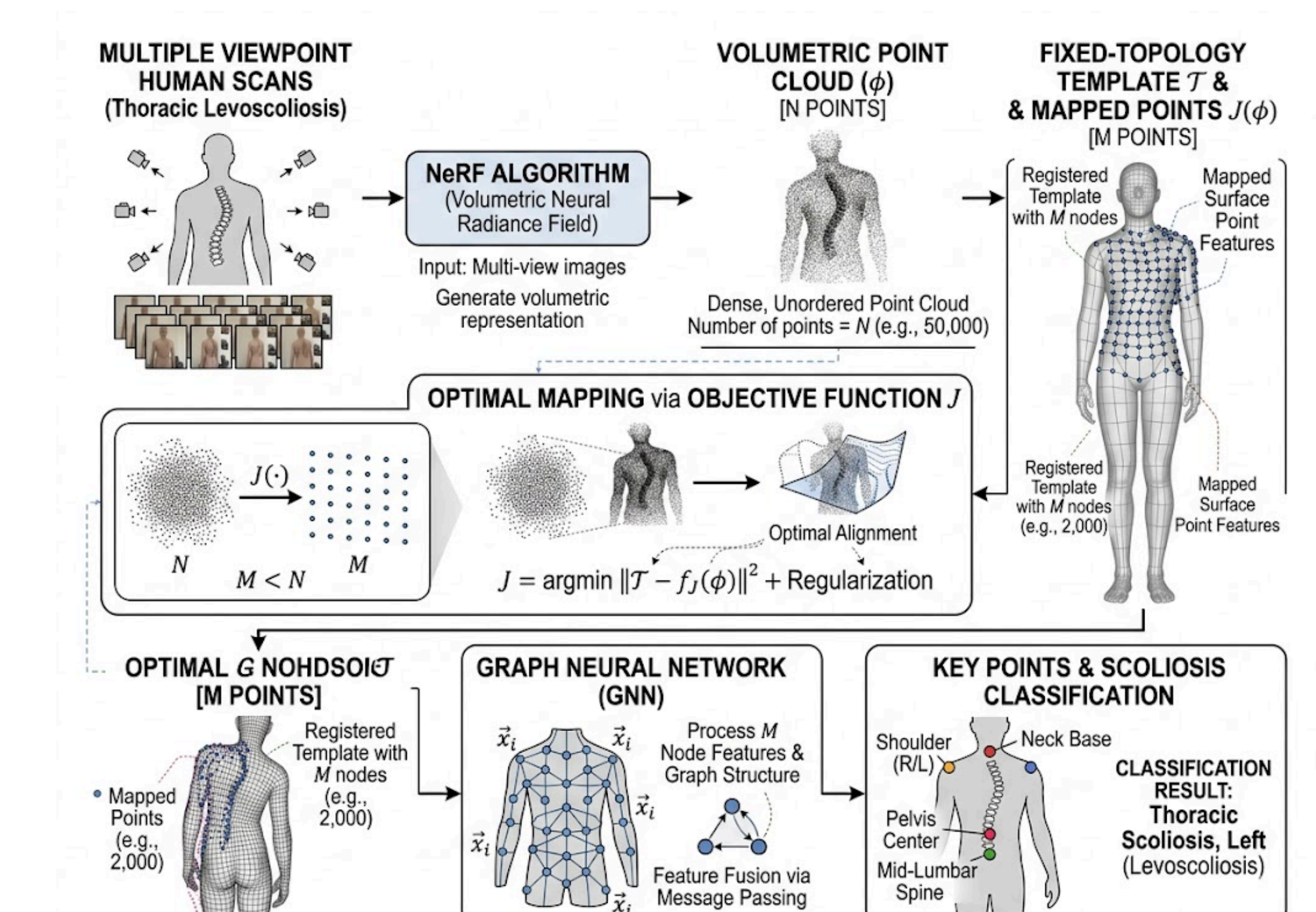
To standardize topology of a volumetric output to a designated number of points (M), we define the problem as an optimization where a **fixed-topology template** is warped to match the zero-level set of the volumetric NeRF field, using the objective function

$$J(\phi) = \int_{\mathcal{R}} \mathcal{L}_{dist}(S_{\phi}(\mathbf{x}), \hat{f}_{NeRF}(\mathbf{x})) d\mathbf{x} + \lambda \mathcal{R}_{smooth}(\phi)$$

Where:

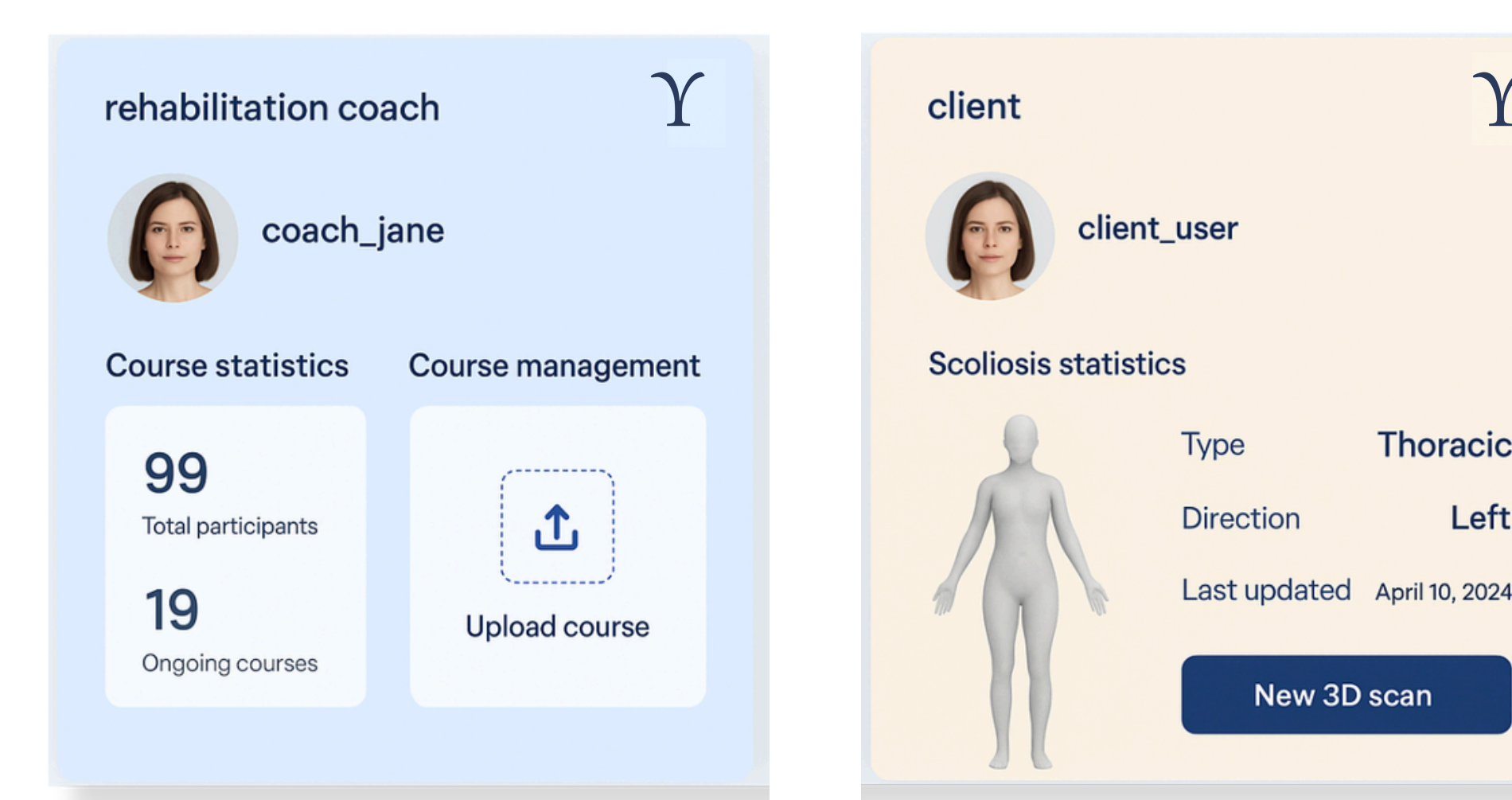
- $\hat{f}_{NeRF}(\mathbf{x})$: The pre-trained NeRF density field.
- $S_{\phi}(\mathbf{x})$: The standardized output generated by a deformation function that transforms a template point set $\mathcal{T}_{fixed}$ into the target shape.
- $\mathcal{L}_{dist}$: A distance metric that minimizes the discrepancy between the NeRF-defined volume and the M points.
- $\mathcal{R}_{smooth}$: An edge-length regularization term to prevent self-intersection and ensure that the standardized points remain a manifold.

With this, it's ensured that the GNN is able to process a point cloud of fixed number of vertices.

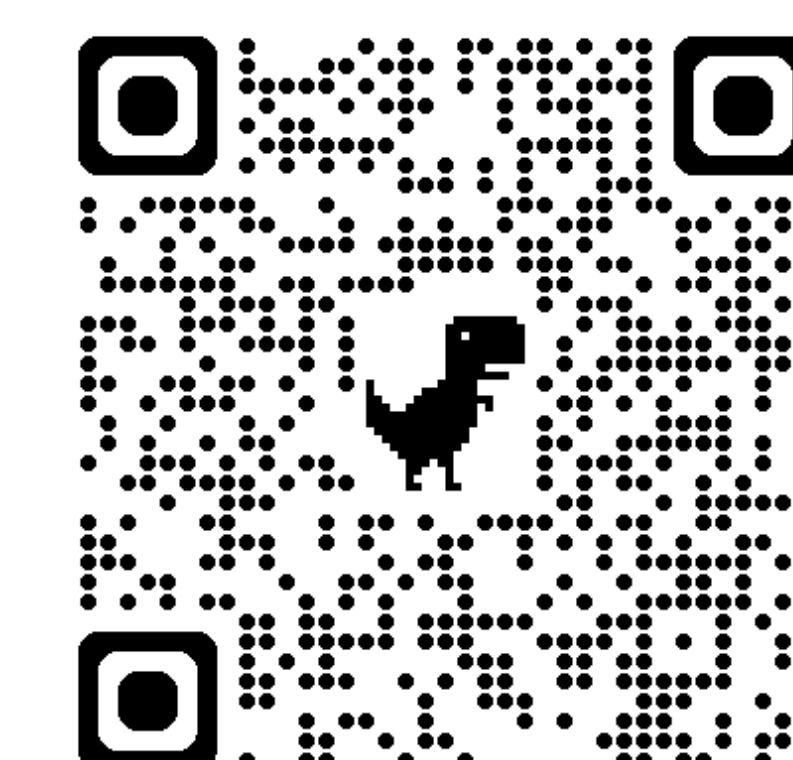


Above is the complete architecture of YXiZ: a multi-view video is **transformed to a density field by NeRF**. The point cloud is then **standardized by optimal mapping** and **processed by GNN to recognize body key points and classify scoliosis**.

Bottleneck: Data Labeling. The Prediction phase is thus yet implemented due to the lack of labeled data. It would be much easier to work with 2D images, but scoliosis often involve spinal rotation, whereby a third dimensional parameter is useful.



Potential UI Interface



GitHub Demo

References

- [1] Weng, C., Curless, B., Srinivasan, P. P., Barron, J. T., & Kemelmacher-Shlizerman, I. (2022, January 11). HumanNeRF: Free-viewpoint Rendering of Moving People from Monocular Video. arXiv.org. <https://arxiv.org/abs/2201.04127>